INCH-POUND
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SUPERSEDING
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DETAIL SPECIFICATION

MOUNTS, RESILIENT: TYPES 6E100, 6E150, 7E450, 6E900, 6E2000, 5E3500, 6E100BB, 6E150BB, 7E450BB, AND 6E900BB

This specification is approved for use by the Naval Sea Systems Command and is available for use by all departments and agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the Naval Ship Research and Development Center rubber resilient mount Types 6E100, 6E150, 7E450, 6E900, 6E2000, 5E3500, 6E100BB, 6E150BB, 7E450BB, and 6E900BB.

1.2 Classification.

1.2.1 Types. The resilient mounts are in accordance with Drawing 803-1385778 and furnished in the following types as specified (see 6.2):

a. 6E100 (load capacity of 50 to 100 pounds)
b. 6E150 (load capacity of 100 to 150 pounds)
c. 7E450 (load capacity of 150 to 450 pounds)
d. 6E900 (load capacity of 400 to 900 pounds)
e. 6E2000 (load capacity of 700 to 2,000 pounds)
f. 5E3500 (load capacity of 2,000 to 3,500 pounds)
g. 6E100BB (load capacity of 50 to 100 pounds)
h. 6E150BB (load capacity of 100 to 150 pounds)
i. 7E450BB (load capacity of 150 to 450 pounds)
j. 6E900BB (load capacity of 400 to 900 pounds)

1.2.1.1 Type nomenclature. The numbers and letters in the type designation denote the following:

a. First number – Denotes the resonance frequency (hertz) in the axial direction (see 6.4.1) at upper rated load
b. Letter E – Denotes the mount design activity (Naval Ship Research and Development Center)
c. Last number – Denotes the upper load rating (pounds)
d. Letters BB – Denote that the mount is designed in a back-to-back configuration

Comments, suggestions, or questions on this document should be addressed to Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-407 - Visual Inspection Guide for Rubber Molded Items

(Copies of this document are available online at http://quicksearch.dla.mil.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

NAVAL SEA SYSTEMS COMMAND (NAVSEA) DRAWINGS

803-1385778 - Mount, Resilient, EES Type

(Copies of this document are available from the applicable repositories listed in S0005-AE-PRO-010/EDM, which can be obtained online via Technical Data Management Information System (TDMIS) at https://mercury.tdmis.navy.mil/. Copies of this document may also be obtained from the Naval Ships Engineering Drawing Repository (NSEDR) online at https://199.208.213.105/webjedmics/index.jsp. To request an NSEDR account for drawing access, send an email to NNSY_JEDMICS_NSEDR_HELP_DESK@navy.mil.)

NAVAL SEA SYSTEMS COMMAND (NAVSEA) PUBLICATIONS

T9070-AL-DPC-020/077-2 - NAVSEA Hazardous Material Avoidance Process

(Copies of this document are available online via Technical Data Management Information System (TDMIS) at https://mercury.tdmis.navy.mil/ by searching for the document number without the suffix. Refer questions, inquiries, or problems to: DSN 296-0669, Commercial (805) 228-0669. This document is available for ordering (hard copy) via the Naval Logistics Library at https://nll.ahf.nmci.navy.mil. For questions regarding the NLL, contact the NLL Customer Service at nllhelpdesk@navy.mil, (866) 817-3130, or (215) 697-2626/DSN 442-2626.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

ASTM INTERNATIONAL

ASTM D395 - Standard Test Methods for Rubber Property – Compression Set

ASTM D412 - Standard Test Methods for Vulcanized Rubber and Thermoplastic Elastomers – Tension

ASTM D429 - Standard Test Methods for Rubber Property – Adhesion to Rigid Substrates

ASTM D471 - Standard Test Method for Rubber Property – Effect of Liquids

ASTM D573 - Standard Test Method for Rubber – Deterioration in an Air Oven

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ASTM D792	-	Standard Test Methods for Density and Specific Gravity (Relative Density) of Plastics by Displacement
ASTM D1005	-	Standard Test Method for Measurement of Dry-Film Thickness of Organic Coatings Using Micrometers
ASTM D1149	-	Standard Test Methods for Rubber Deterioration – Cracking in an Ozone Controlled Environment
ASTM D1229	-	Standard Test Method for Rubber Property – Compression Set at Low Temperatures
ASTM D2240	-	Standard Test Method for Rubber Property – Durometer Hardness
ASTM D5992	-	Standard Guide for Dynamic Testing of Vulcanized Rubber and Rubber-Like Materials Using Vibratory Methods
ASTM D7091	-	Standard Practice for Nondestructive Measurement of Dry Film Thickness of Nonmagnetic Coatings Applied to Ferrous Metals and Nonmagnetic, Nonconductive Coatings Applied to Non-Ferrous Metals

(Copies of these documents are available online at www.astm.org.)

INTERNATIONAL ORGANIZATION FOR STANDARDIZATION (ISO)

ISO 10846-1	-	Acoustics and Vibration – Laboratory Measurement of Vibro-Acoustic Transfer Properties of Resilient Elements - Part 1: Principles and Guidelines
ISO 10846-2	-	Acoustics and Vibration – Laboratory Measurement of Vibro-Acoustic Transfer Properties of Resilient Elements - Part 2: Direct Method for Determination of the Dynamic Stiffness of Resilient Supports for Translatory Motion

(Copies of these documents are available online at www.iso.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. Mounts furnished under this specification shall be products that are authorized by the qualifying activity for listing on the applicable qualified products list before contract award (see 4.2 and 6.3).

3.2 Materials.

3.2.1 Metal. The metal components of the mount shall be manufactured from steel or manganese bronze as specified (see 6.2), in accordance with Drawing 803-1385778. The metal components for the mounts shall be formed to shape and finished in accordance with the dimensions and allowable tolerances specified by the mount drawing.

3.2.2 Rubber. The resilient elements of the 7E450, 6E900, 6E2000, 7E450BB, and 6E900BB mounts shall be fabricated of oil-resistant compounds utilizing polymerized chloroprene as the basic material. The resilient elements of the 6E100, 6E150, 5E3500, 6E100BB, and 6E150BB mounts shall be molded from compounds using natural rubber as the basic material. The natural rubber components shall be coated with an oil-ozone resistant compound. Coating of the mounts may be accomplished by, but is not limited to, dipping, painting, or spraying. Refer to 6.6. The mounts shall be molded to the specified form as shown on Drawing 803-1385778.

3.2.2.1 Porosity and delamination. The rubber elements of the furnished mounts shall have no delamination, air pockets, or porosity in any portion when visually examined or tested in accordance with 4.4.18.

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3.2.2.2 Bond. The rubber elements shall be bonded to the metal components as specified on Drawing 803-1385778.

3.2.2.3 Surface condition. There shall be no backrinding, blisters, or other defects on the outer surfaces of the rubber elements of the finished mounts. Refer to table XIV.

3.2.2.4 Physical requirements.

3.2.2.4.1 Rubber compounds. The vulcanized rubber compounds used in the resilient elements of the mounts shall be in accordance with the physical requirements specified in table I.

3.2.2.4.2 Hardness. The hardness of the rubber compounds used in the resilient elements shall be determined in accordance with 4.4.10. The hardness found during qualification inspection of the mount compounds shall be recorded as a requirement with a tolerance of ±5 points for subsequent conformance testing of the mount compounds.

3.2.2.4.3 Specific gravity. The specific gravity of the rubber compounds used for the resilient elements shall be determined in accordance with 4.4.11. The specific gravity found during qualification inspection of the mount compounds shall be recorded as a requirement with a tolerance of ±0.02 for subsequent conformance testing of the mount compounds.

TABLE I. Physical requirements for rubber compounds.

	Resilient elements				**Test method**
	Type 6E100, 6E100BB mounts	**Type 6E150, 5E3500, 6E150BB mounts**	**Type 7E450BB mounts**	**Type 6E900, 6E900BB, 6E2000 mounts**	
Tensile strength (lb/in^2, min.)					
Before aging	2,200	3,200	2,200	2,500	4.4.2
After aging 46 hours at 194 °F	2,100	2,600	2,100	2,400	4.4.4
Elongation at break (min.), percent					
Before aging	550	500	600	600	4.4.2
After aging 46 hours at 194 °F	500	450	550	550	4.4.4
Compression set (max.), percent after aging 46 hours at 194 °F	30	35	37	37	4.4.3.1
Cold compression set (max.), percent	20	20	50	60	4.4.3.2
Volume change in oil (max.), percent	---	---	15	15	4.4.5.1
Adhesion of rubber-to-metal (lb/in^2, min.)	400	500	450	500	4.4.6.1

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3.2.2.4.4 Oil-ozone resistant coating. The oil-ozone resistant coating shall be in accordance with the physical requirements specified in table II (see 6.6).

TABLE II. Physical requirements for oil-ozone resistant coating.

Property	Requirement	Test
Properties of dried film:		
Tensile strength (lb/in^2, min.)	2,000	4.4.2
Elongation at break (min.), percent	400	4.4.2
Properties of coated test specimens:		
Volume change (max.) after immersion in oil at 158 °F for 70 hours, percent	5	4.4.5.2
Adhesion of coating:		
Before immersion in oil	No failures	4.4.6.2
After immersion in oil at 158 °F	No failures	4.4.6.2
Ozone resistance after 1 week at 104 °F with ozone partial pressure of 100 millipascals (mPa)	No cracks	4.4.7
Flexibility of coating after flexing for 6 cycles to 100 percent elongation	No cracks	4.4.8
Properties of coated mount:		
Appearance	Shall be dry, tack free, and free from blisters, imperfections, or other defects listed in table XIV	table XIV
Film thickness (min.) on rubber element as measured on coated metal, inch	0.003	4.4.9

3.2.3 Protective treatment. The metal components and elastomers used in the mounts shall be resistant to or protected against corrosion by seawater, oil, ozone, or other atmospheric conditions encountered in service. Unless otherwise specified (see 6.2), the minimum protective treatment shall be equivalent to that specified on Drawing 803-1385778. Unless otherwise specified (see 6.2), the coating on the metal shall be equivalent to that specified on the drawing and shall have a minimum dry film thickness of 0.006 inch.

3.2.4 Toxicity and prohibited materials.

3.2.4.1 Toxicity. When evaluated in accordance with 4.5, the mount shall pose no serious or high risk to the health of personnel or the environment when used for its intended purpose (see 4.5 and 6.9).

3.2.4.2 Prohibited materials. The mount shall not contain any chemicals categorized as “prohibited” in accordance with T9070-AL-DPC-020/077-2.

3.2.5 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

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3.3 Requirements for the finished mounts.

3.3.1 Dynamic stiffness. Dynamic stiffness of the mounts shall be within the specified limits of table III when tested in the axial direction in accordance with 4.4.12.

TABLE III. Axial dynamic stiffness requirements at various loads.

Mount	Requirement-limits		
	Test load (lb)	Calculated resonant frequency (Hz)	Dynamic stiffness (lb/in)
6E100	50	7.6 – 8.8	294 – 395
	70	6.2 – 7.5	274 – 402
	85	5.6 – 6.8	271 – 401
	100	5.3 – 6.6	286 – 444
6E150	100	5.7 – 7.7	331 – 605
	120	5.2 – 7.1	330 – 617
	135	4.7 – 6.7	304 – 618
	150	4.4 – 6.3	296 – 607
7E450	150	7.2 – 9.1	793 – 1,267
	250	6.2 – 8.2	980 – 1,715
	350	6.1 – 7.7	1,310 – 2,123
	450	6.0 – 7.5	1,625 – 2,582
6E900	400	5.0 – 7.0	1,020 – 1,999
	540	5.0 – 7.0	1,377 – 2,699
	720	5.0 – 7.0	1,836 – 3,599
	900	5.0 – 7.0	2,295 – 4,498
6E2000	700	5.0 – 7.0	1,785 – 3,499
	1,150	5.0 – 7.0	2,932 – 5,748
	1,600	5.0 – 7.0	4,080 – 7,997
	2,000	5.0 – 7.0	5,100 – 9,996
5E3500	2,000	4.0 – 6.1	3,264 – 7,500
	2,500	4.0 – 6.1	4,080 – 9,500
	3,000	4.0 – 6.1	4,896 – 11,500
	3,500	4.0 – 6.1	5,712 – 13,500
6E100BB	50	7.6 – 10.2	294 – 531
	70	6.6 – 9.0	311 – 578
	85	5.9 – 7.9	301 – 541
	100	5.6 – 7.6	319 – 589
6E150BB	100	6.9 – 9.3	485 – 882
	120	6.3 – 8.6	485 – 905
	135	6.0 – 8.2	495 – 925
	150	5.7 – 7.7	497 – 907

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TABLE III. Axial dynamic stiffness requirements at various loads – Continued.

Mount	Requirement-limits		
	Test load (lb)	Calculated resonant frequency (Hz)	Dynamic stiffness (lb/in)
74E450BB	150	6.5 – 9.1	646 – 1,267
	250	6.5 – 8.5	1,077 – 1,842
	350	6.5 – 8.5	1,487 – 2,610
	450	6.5 – 8.5	1,939 – 3,316
6E900BB	400	5.0 – 7.5	1,020 – 2,295
	540	5.0 – 7.5	1,377 – 3,098
	720	5.0 – 7.5	1,836 – 4,131
	900	5.0 – 7.5	2,295 – 5,164

3.3.2 Deflection at rated load. Mount deflections at their respective rated loads shall be within the specified limits of table IV when tested in the axial direction in accordance with 4.4.13. The mount shall not show any damage, break, or separation in or between the component parts.

TABLE IV. Deflection at rated load.

Mount	Test load	Deflection (inch)	
		Minimum	Maximum
6E100	100	0.20	0.35
6E150	150	0.22	0.37
7E450	450	0.26	0.44
6E900	900	0.37	0.62
6E2000	2,000	0.40	0.68
5E3500	3,500	0.46	0.64
6E100BB	100	0.34	0.56
6E150BB	150	0.25	0.42
7E450BB	450	0.41	0.69
6E900BB	900	0.52	0.88

3.3.3 Quality of the rubber-to-metal bond. Mounts shall show no damage, breaks, cracks, or tears in the rubber elements or evidence of delamination at rubber-to-metal bond interfaces when tested in accordance with 4.4.14.

3.3.4 Strength. Mounts, when tested in the axial direction in accordance with 4.4.15.1 or in the radial direction (see 6.4.2) in accordance with 4.4.15.2, shall not show any damage, break, or separation in the rubber or between rubber and metal parts, or any permanent deformation of the metal parts in excess of 1/32 inch.

3.3.5 Drift. Mounts shall not exceed the drift requirements of table V when tested in accordance with 4.4.16. There shall be no failure of the resilient element, rubber-to-metal bond, or metal parts during the tests. At the end of testing, the dynamic stiffness of each mount shall meet the requirements specified in table III.

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TABLE V. Drift requirements.

Mount	Drift (max.) inch
6E100	0.025
6E150	0.025
7E450	0.050
6E900	0.060
6E2000	0.080
5E3500	0.035
6E100BB	0.025
6E150BB	0.025
7E450BB	0.050
6E900BB	0.060

3.3.6 Fatigue. Mounts shall withstand 500,000 cycles of fatigue loading without any signs of failure or deterioration in the resilient element, rubber-to-metal bond, oil-ozone resistant coating, or metal components when tested in accordance with 4.4.17.

3.3.7 Porosity and delamination. The resilient elements of the mounts shall show no evidence of porosity in the rubber or separation of the rubber into distinct layers or laminations when tested and examined in accordance with 4.4.18.

3.3.8 Mount design. Each mount shall be in accordance with the applicable details shown on Drawing 803-1385778. Each mount shall be identified with the applicable markings specified on the drawing. There shall be no defects (see table XIV) when examined.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 Qualification inspection. Qualification inspection shall include the examination and tests contained in 4.2.1 and 4.2.2.

4.2.1 Rubber samples and tests for qualification inspection.

4.2.1.1 Rubber samples. For the tests specified in 4.2.1.2 and 4.2.1.3, the samples listed in table VI shall be provided.

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TABLE VI. Rubber samples for qualification.

Quantity	Type of rubber compound	Sample description	Size of sample (inches)	Property to be measured
3	Resilient element	Sheet	0.08±0.01 × 6 × 6	Tensile strength and ultimate elongation (initial and aged) Specific gravity
3	Resilient element	Sheet 1/	0.08±0.01 × 1 × 2	Resistance to oil, volume change
3	Resilient element	Specimen covered with oil-ozone resistant coating 2/	0.08±0.01 × 1 × 2	Resistance to oil, volume change
3	Resilient element	Specimen covered with oil-ozone resistant coating 2/	0.08±0.01 × 1 × 6	Ozone resistance, adhesion and flexibility of coating
3	Resilient element	ASTM D429 Method A specimen	---	Rubber-metal adhesion
4	Resilient element	Cylinder	0.50±0.02 height × 1.14±0.02 diameter	Compression set and hardness
1	Oil-ozone resistant coating	Sheet	0.006±0.003 × 6 × 6	Tensile strength and ultimate elongation

NOTES:

1/ Uncoated resilient element sheet specimens for 7E450, 6E900, 6E2000, 7E450BB, and 6E900BB mounts only.

2/ Resilient element specimens covered with oil-ozone resistant coating for 6E100, 6E150, 5E3500, 6E100BB, and 6E150BB mounts only.

4.2.1.2 Uncoated rubber samples. The uncoated rubber samples specified in 4.2.1.1 shall be subjected to the tests specified in table VII.

TABLE VII. Tests on uncoated rubber samples.

Property	Requirement	Inspection
Initial tensile strength	table I	4.4.2
Initial ultimate elongation	table I	4.4.2
Tensile strength after oven aging	table I	4.4.2 and 4.4.4
Ultimate elongation after oven aging	table I	4.4.2 and 4.4.4
Compression set after oven aging	table I	4.4.3.1
Cold compression set	table I	4.4.3.2
Resistance to oil 1/	table I	4.4.5.1
Adhesion to metal	table I	4.4.6.1
Hardness	3.2.2.4.2	4.4.10
Specific gravity	3.2.2.4.3	4.4.11

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TABLE VII. Tests on uncoated rubber samples – Continued.

NOTE:
1/ 7E450, 6E900, 6E2000, 7E450BB, and 6E900BB mounts only.

4.2.1.3 Oil-ozone resistant coating samples. The oil-ozone resistant coating and coated rubber samples specified in 4.2.1.1 shall be subjected to the tests specified in table VIII.

TABLE VIII. Tests on coated rubber and dried film samples.

Test	Requirement	Inspection
Tensile strength of dried film	table II	4.4.2
Ultimate elongation of dried film	table II	4.4.2
Swelling of coated rubber sample 1/	table II	4.4.5.2
Adhesion of coating 1/	table II	4.4.6.2
Ozone resistance 1/	table II	4.4.7
Flexibility of coating 1/	table II	4.4.8
NOTE: 1/ 6E100, 6E150, 5E3500, 6E100BB, and 6E150BB mounts only.		

4.2.2 Qualification inspection on finished mounts.

4.2.2.1 Mount samples. If the rubber samples listed in table VI meet all requirements in 4.2.1, the tests specified in 4.2.2.2 and 4.2.2.3 shall be conducted using four mounts of each type manufactured from the compounds represented by the rubber samples.

4.2.2.2 Group one. For each mount type, two of the four mounts specified in 4.2.2.1 shall be subjected to the tests in table IX in the sequence shown.

TABLE IX. Test group one.

Test	Requirement	Inspection
Examination	No defects	table XIV
Film thickness of oil-ozone resistant coating	table II	4.4.9
Dynamic stiffness	3.3.1	4.4.12
Deflection at rated load	3.3.2	4.4.13
Quality of rubber-to-metal bond	3.3.3	4.4.14
Strength, axial	3.3.4	4.4.15.1
Strength, radial	3.3.4	4.4.15.2

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4.2.2.3 Group two. For each mount type, the remaining two mounts specified in 4.2.2.1 shall be subjected to the tests in table X in the sequence shown.

TABLE X. Test group two.

Test	Requirement	Inspection
Examination	No defects	table XIV
Dynamic stiffness	3.3.1	4.4.12
Deflection at rated load	3.3.2	4.4.13
Quality of rubber-to-metal bond	3.3.3	4.4.14
Drift	3.3.5	4.4.16
Fatigue	3.3.6	4.4.17
Porosity and delamination	3.3.7	4.4.18

4.3 Conformance inspection. Conformance inspection shall be performed on each production lot and include the examination and tests contained in 4.3.2 and 4.3.3.

4.3.1 Lot. For the purpose of conformance inspection and sampling, a production lot shall consist of all the mounts of one type, design, and load rating produced in the same facility under the same conditions with the same materials being offered for delivery at one time. A lot serial number shall be assigned to each mount. The lot serial number shall not be repeated in any one quarter. The serial number shall be traceable to the mount rubber batch number, the manufacturing and process records against which the mounts were manufactured, and all conformance requirements as required by this specification.

4.3.2 Rubber samples and tests for conformance inspection.

4.3.2.1 Rubber compound samples. As specified in table XI, samples shall be provided from each batch of rubber stock that is mixed for manufacture of the mounts in the lot. The samples shall be certified to be of the same material and equivalent cure as the corresponding components in the lot of finished mounts offered for delivery.

TABLE XI. Rubber samples for conformance inspection.

Quantity	Type of rubber compound	Description of sample	Size of sample (inch)	Property to be measured
2	Resilient element	Sheet	0.08±0.01 × 6 × 6	Tensile strength and ultimate elongation specific gravity
1	Resilient element	Slab or disc	0.5 thick	Hardness

4.3.2.2 Conformance tests on rubber samples. Samples as specified in 4.3.2.1 shall be subjected to the tests specified in table XII.

TABLE XII. Conformance tests on rubber samples.

Test	Requirement	Inspection
Tensile strength and ultimate elongation	table I	4.4.2
Hardness	3.2.2.4.2	4.4.10
Specific gravity	3.2.2.4.3	4.4.11

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4.3.2.3 Rejection. If any samples tested in accordance with 4.3.2.2 fail to meet the requirements, all mounts in the lot represented by the samples shall be rejected.

4.3.3 Conformance inspection on mounts.

4.3.3.1 Visual and dimensional examination.

4.3.3.1.1 Sampling for visual and dimensional examination. As a minimum, a sample quantity of mounts from each lot shall be randomly selected in accordance with table XIII and inspected in accordance with 4.3.3.1.2. Sample size depends on the defect category. If one or more defects are found in any sample, the defective mounts shall be rejected and the entire lot screened for all defects. All defective mounts shall be rejected and removed from the lot.

TABLE XIII. Sampling for visual and dimensional examination.

Lot size	Sample size	
	Minor defects	Major defects
90 and under	5	8
91 to 280	7	12
281 to 500	9	16
501 to 1,200	11	19
1,201 to 3,200	13	23

4.3.3.1.2 Procedure for visual and dimensional examination. Each of the mounts selected in accordance with table XIII shall be visually and dimensionally examined for the defects listed in table XIV to verify compliance with this specification. MIL-STD-407 shall be used to determine and evaluate defects through visual examination.

4.3.3.2 Conformance tests on mounts.

4.3.3.2.1 Sampling for tests. As a minimum, a sample quantity of mounts from each lot shall be selected in accordance with table XV, and each mount in the sample shall be subjected to the tests specified in 4.3.3.2.2. If the quality of rubber-to-metal bond requirements have not been met by all mounts in the sample, the entire lot shall be rejected. If dynamic stiffness, deflection at rated load, and thickness of oil-ozone resistant film requirements have not been met by all mounts in the sample, the defective mount(s) shall be rejected, and the entire lot shall be tested for the defect(s) found. All defective mounts shall be rejected and removed from the lot.

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TABLE XIV. Classification of physical defects.

Category	Defects
Major:	
101	Evidence of use of unauthorized materials
102	Rubber elements not molded to specified form on drawing
103	Evidence of delamination or air pockets in rubber elements of finished product
104	Rubber elements not bonded to metal components in accordance with drawing
105	Evidence of backrinding, blisters, cracks, breaks, gouges, tears, or other imperfections on the surface of rubber elements.
106	Evidence of tackiness, non-drying, pealing, or other imperfections in the oil-ozone resistant coating. Required coating absent.
107	Metal components not protected from corrosion by seawater or spray or other atmospheric conditions encountered in service
108	Dimensions, configuration, and construction not in accordance with drawing
109	Any other defect that would affect the serviceability, performance, or service life of the mounts
Minor:	
201	Identification marking not distinct
202	Burrs, rough edges, and sharp corners not removed
203	Any other defect that would not affect the serviceability, performance, or service life of the mounts

TABLE XV. Sampling for conformance tests.

Lot size	Sample size
15 and under	5
16 to 25	7
26 to 90	8
91 to 150	13
151 to 280	20
281 to 500	29
501 to 1,200	34
1,201 to 3,200	42

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4.3.3.2.2 Tests to be performed on mount samples. Each of the samples selected in accordance with 4.3.3.2.1 shall be subjected to the tests specified in table XVI.

TABLE XVI. Conformance tests.

Test [1/]	Requirement	Test
Thickness of oil-ozone resistant film	table II	4.4.9
Dynamic stiffness	3.3.1	4.4.12
Deflection at rated load	3.3.2	4.4.13
Quality of rubber-to-metal bond	3.3.3	4.4.14
NOTE: 1/ Listed in sequence of performance.		

4.4 Test methods.

4.4.1 Standard test conditions. Unless otherwise specified in the test procedure, all tests shall be made within an ambient temperature range of 65 to 89 °F.

4.4.2 Tensile and ultimate elongation tests. Tensile strength and ultimate elongation tests shall be conducted on the resilient element rubber compounds and the 0.006-inch thick coating film in accordance with ASTM D412. Die C test specimens shall be used. Tensile strength and elongations of the rubber compounds and the coatings shall meet the requirements of tables I and II.

4.4.3 Compression set.

4.4.3.1 Compression set after aging. Compression set shall be determined in accordance with ASTM D395 Method B, except oven aging conditions shall be 194±2 °F for 46±¼ hours.

4.4.3.2 Cold compression set. Cold compression set shall be determined in accordance with ASTM D1229, except the exposure shall be 30±2 °F for 94±½ hours. Compression set shall be determined 30 minutes after release from compression.

4.4.4 Oven aging test. Specimens for tensile, ultimate elongation, and compression set tests shall be oven aged in accordance with ASTM D573 at a temperature of 194±2 °F for 46±¼ hours. Final determination of aged tensile and ultimate elongation specimens shall be made not less than 16 hours or more than 48 hours after removal from the oven. Tensile and ultimate elongation tests on unaged specimens shall be made immediately prior to, and on the same machine as, the tensile tests on the oven aged specimens (see 4.4.2).

4.4.5 Swelling in oil.

4.4.5.1 Volume change in oil (uncoated rubber compounds). Volume change shall be determined in accordance with ASTM D471 on uncoated 0.08- by 1- by 2-inch specimens as described in table VI. The specimen shall be immersed in IRM 903 reference oil in accordance with ASTM D471. The immersion period shall be 46±¼ hours at 73±2 °F.

4.4.5.2 Volume change in oil (coated samples). Volume change in oil shall be determined in accordance with ASTM D471 on the three coated 0.08- by 1- by 2-inch specimens as described in table VI. The specimen shall be immersed in IRM 903 reference oil in accordance with ASTM D471. The immersion period shall be 70±¼ hours at 158±2 °F.

4.4.6 Adhesion tests.

4.4.6.1 Adhesion to metal tests. The adhesion tests shall be in accordance with Method A of ASTM D429. As described in table VI, three specimens of each rubber compound shall be tested and the results averaged.

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4.4.6.2 Adhesion of coating (coated sample). The adhesion of the oil-ozone resistant coating to three 0.08- by 1- by 2-inch samples of the coated rubber specimens (see table VI) shall be determined before and after immersion in oil in accordance with 4.4.5.2. Each coated specimen shall be flexed, elongated by hand, and then visually examined for adhesion failures. The coated specimens shall not exhibit cracks, breaks, tears, or blisters conducive to peeling of the coating by hand either before or after immersion in oil.

4.4.7 Ozone resistance. The ozone resistance of the coated rubber components of the mounts shall be determined on two coated 1- by 6- by 0.080-inch thick specimens (see table VI). The specimens shall be elongated to 20 percent extension. A thin layer of melted paraffin wax shall be applied to each of the four surfaces of the stretched specimens in an area not exceeding a ¼-inch width from the clamps. The stretched specimens shall be conditioned for 16±2 hours at 104±2 °F before being exposed, in accordance with ASTM D1149. The standard ozone partial pressure shall be 100 millipascals (mPa) ±10 percent, the temperature shall be 104±2 °F, and the period of exposure shall be 168 hours. Observation magnification shall be 7X.

4.4.8 Flexibility of the coating. The flexibility of the coated 1- by 6- by 0.080-inch coated rubber specimens specified in table VI shall be determined. Each specimen shall be clamped in a universal test machine and a tensile force applied to elongate the specimen 100 percent at a loading rate of 20 inches per minute. The force shall then be released so that the specimen is no longer elongated. The cycle shall be repeated six times. The coated specimens shall be visually examined and shall exhibit no cracks, breaks, tears, or blisters.

4.4.9 Film thickness of the oil-ozone resistant coating. The film thickness of the oil-ozone resistant coating as specified in table II shall be measured on the coated steel plates of the mount using a micrometer. The thickness of the steel plate shall be determined before coating and after 0.003 inch of multiple coatings have been applied to the mount. The difference in the two thickness measurements divided by two shall be the film thickness of the oil-ozone resistant coating. Refer to ASTM D1005. In lieu of the above procedure, the film thickness may also be measured by using a magnetic type thickness gauge in accordance with ASTM D7091.

4.4.10 Hardness. The hardness of the 0.50-inch thick specimens representing the rubber components of the mount shall be determined in accordance with ASTM D2240. A 3-second reading shall be taken to determine conformance with 3.2.2.4.2.

4.4.11 Specific gravity. The specific gravity of the rubber compounds used for the mounts shall be determined on specimens cut from 0.08- by 6- by 6-inch sheets in accordance with ASTM D792. The average of three determinations shall be recorded to determine conformance with 3.2.2.4.3.

4.4.12 Dynamic stiffness test. Dynamic stiffness of each mount shall be measured in the axial direction at the static loads specified in table XVII to determine conformance with 3.3.1. Type 6E100BB, 6E150BB, 7E450BB, and 6E900BB mounts shall be tested in the zero preload condition. Prior to conducting dynamic stiffness tests, all mounts shall first be “conditioned” a minimum of three cycles to the maximum test load in table XVII below. The general guidelines and considerations for dynamic stiffness testing given in ASTM D5992, ISO 10846-1, and ISO 10846-2 shall be followed.

TABLE XVII. Loads for testing dynamic stiffness.

Mount type	Loads (pounds)
6E100, 6E100BB	50, 70, 85, and 100
6E150, 6E150BB	100, 120, 135, and 150
7E450, 7E450BB	150, 250, 350, and 450
6E900, 6E900BB	400, 540, 720, and 900
6E2000	700, 1,150, 1,600, and 2,000
5E3500	2,000, 2,500, 3,000, and 3,500

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4.4.12.1 Test system for measuring dynamic stiffness. The test system shall be capable of superimposing cyclical vibration while statically loading the mount. Refer to ASTM D5992 for examples of non-resonant forced vibration test systems and guidance in addressing mechanical and instrumentation factors that influence dynamic measurements. A commonly used type of equipment for non-resonant testing is a closed-loop servo-hydraulic machine (see figure 1). Transducers shall be arranged across the mount to measure force at the blocked output side of the mount (foundation) and displacement at the input side of the mount (equipment). The stiffness of the test apparatus (machine, load-frame, force transducer, etc.) in series with the mount under test shall be determined experimentally (see 6.5) or via calculation. A test apparatus that is at least 100 times stiffer than the dynamic stiffness of the mount being tested shall not require measurement adjustment to account for the flexibility of the test apparatus. If the test apparatus fails to meet this relative stiffness requirement, the test apparatus compliance shall be subtracted from the measurement to obtain the dynamic stiffness of the mount (K_{mount}). The dynamic stiffness of the mount can be calculated using the classic equation governing the summation of springs in series:

$$(1/K_{measured}) = (1/K_{mount}) + (1/K_{apparatus}) \qquad \text{yielding} \qquad K_{mount} = \frac{(K_{apparatus})(K_{measured})}{(K_{apparatus}) - (K_{measured})}$$

To avoid over-correction, the stiffness of the test apparatus shall be greater than 30 times the stiffness of the mount.

4.4.12.2 Confirmation of the measurement system. Refer to ASTM D5992 for mechanical and instrumentation factors influencing dynamic measurement, selection, and use of springs to assess the capability of the test apparatus to measure dynamic stiffness. Prior to conducting dynamic stiffness measurements on mounts, the dynamic calibration of the complete system shall be confirmed using steel springs of known spring constants, having negligible damping. Springs shall be free of hysteresis and their dynamic and static stiffness (tangent to the slope of the load-deflection curve) equivalent. This procedure is not intended to replace the periodic calibration of transducers, signal conditioners, and other measuring instruments. Springs shall be selected on the basis of their linear force-deflection properties and ability to produce a spring constant comparable to the stiffness of the mount to be tested. Refer to ASTM D5992 for factors influencing the selection of springs. System confirmation shall be performed using springs of known spring constants at the excitation amplitudes and frequencies at which the mount is to be tested. System confirmation shall be performed at loads that produce spring constant measurements that bracket the expected dynamic stiffness range of the mount to be tested. This shall be accomplished by adding or removing springs within the stiffness range of intended mount measurements. Care shall be taken to ensure measurements are conducted within the linear load range of the springs. All spring dynamic stiffness measurements (corrected for test apparatus stiffness) should be within 2 percent of the known spring constants. If this is not the case, the error shall be investigated and corrected. A system scaling factor may only be applied to measured dynamic stiffness with NAVSEA approval.

4.4.12.3 Measuring mount dynamic stiffness. Dynamic stiffness shall be measured in the axial direction at each of the static test loads in table XVII. The nominal excitation frequency shall be at 6 Hertz for all mounts except the 7E450, 7E450BB shall be 7 Hertz, and the 5E3500 shall be 5 Hertz. The displacement amplitude of the sinusoidal excitation shall be approximately 0.020 inch peak-to-peak. Dynamic stiffness shall be measured at the excitation frequency by using Fast Fourier Transform (FFT) based methods. This requires measurement of the sinusoidal displacement at the input of the mount and transmitted dynamic force at the blocked output of the mount, as well as the phase relationship between them. It should be noted that damping loss factor is the tangent of the phase between the displacement and force. Dynamic stiffness is calculated from the measurements as follows:

$$K_{measured} = (F/X)(\cos(\theta))$$

Where (F) and (X) are the amplitudes of sinusoidal force and displacement respectively and (θ) is the phase between them. Loss factor (η) can be calculated from the measurements as follows:

$$\eta = \tan(\theta)$$

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In order to obtain mount stiffness (K_{mount}) to determine compliance with 3.3.1, the measured stiffness ($K_{measured}$) may likely require adjustment to account for test apparatus flexibility (see 4.4.12.1).

$$K_{mount} = \frac{(K_{apparatus})(K_{measured})}{(K_{apparatus}) - (K_{measured})}$$

A system scaling or multiplication factor (C) applied to the measured dynamic stiffness ($C*K_{measured}$) is assumed to be unity (C=1) and not shown in the above equation.

4.4.13 Deflection at rated load. A universal type testing machine shall be used. Each mount shall be subjected to a single loading cycle in the axial direction. Types 6E100BB, 6E150BB, 7E450BB, and 6E900BB shall be tested in the zero preload condition. The loading shall be accomplished at a constant rate such that the maximum load is attained in not less than 1 minute. Data shall be corrected for error resulting from flexibility in the test apparatus. Refer to 6.5. The deflection at rated load shall be measured to the closest 0.001 inch. The highest load given in table XVII for each mount is the upper rated load. The deflection of the upper rated load shall be recorded to determine conformance with 3.3.2.

4.4.14 Quality of the rubber-to-metal bond. The load on the mounts, tested in accordance with 4.4.13, shall be increased to 2 times the upper rated load after recording the deflection at the upper rated load. The rate of loading shall be such that the maximum load is attained in not less than 2 minutes. The mount shall be held at 2 times its upper rated load while the appearance of the resilient element and of the rubber-to-metal bond is examined to determine conformance to 3.3.3.

4.4.15 Strength tests. A universal type testing machine shall be employed to conduct strength tests on mounts. The testing machine shall load the mounts at a constant rate of deflection. Data shall be corrected for error resulting from flexibility in the test apparatus. Refer to 6.5. The mounts shall be examined during and after testing for any damage, break, or separation in the rubber or between rubber and metal parts, and deformation of metal parts.

4.4.15.1 Strength tests in the axial direction. The mount shall be secured in a test fixture and subjected to four loading and unloading cycles in the axial direction. Type 6B100BB, 6E150BB, 7E450BB, and 6E900BB mounts shall be tested in the zero preload condition. The loadings shall be conducted at a rate not to exceed 0.3 inch per minute. For the first three loading cycles, the loads shall be applied up to the load specified on curve number 2 of figure 2 for the upper load rating of the mount (35,000 pounds for the 5E3500). For the fourth loading cycle, the mount shall be loaded up to the peak load specified on curve number 1 of figure 2 for the upper load rating of the mount (55,000 pounds for the 5E3500). The deflection at the rated load for each mount shall be noted during the fourth cycle.

4.4.15.2 Strength tests in the radial direction. Types 6E100BB, 6E150BB, 7E450BB, and 6E900BB shall not be tested in the radial direction. Two mounts shall be secured in a fixture similar to that on figure 3 and tested for strength in the radial direction. Each mount shall be compressed axially to a deflection corresponding to its upper rated load (see 4.4.13). This compression is obtained by adjusting the nuts on the four rods, which changes the distance between the brackets and the plane of the loading bar. After assembly, the two mounts shall be subjected to four radial loading and unloading cycles in accordance with the procedure of 4.4.15.1.

4.4.16 Drift test. Each mount shall be loaded to its upper rated load capacity (see 1.2.1) in the axial direction after being previously tested for dynamic stiffness. Types 6E100BB, 6E150BB, 7E450BB, and 6E900BB mounts shall be tested in the zero preload condition. The height of the mount shall be measured to the nearest 0.001 inch 1 hour after loading and again after 96 hours. The difference in the two readings shall be taken as the drift of the mount. Within 15 minutes after the final reading has been obtained, the mount shall be tested again for dynamic stiffness at its upper rated load. Tests shall be conducted at an ambient temperature of 80+5 °F. Temperature shall not vary by more than ±2 °F during the drift test.

4.4.17 Fatigue test. Each mount shall be statically loaded to its upper rated load (see 1.2.1) in the axial direction; the back-to-back mounts shall be tested in the zero preload condition. Each mount shall then be subjected to 500,000 cycles of vibration at its natural frequency; the exciting force shall produce a mount deflection of ±0.050 inch.

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4.4.18 Porosity and delamination. One of the mounts used for the drift test shall be cut into two parts along a vertical plane through the center of the mount. The cut surfaces of these components shall be carefully examined for porosity. The parts shall be immersed in toluene (technical grade) for 24 hours. After removal from the toluene, the rubber elements shall be examined for evidence of separation into distinct layers or laminations.

4.5 Toxicity and prohibited materials. A Health Hazard Assessment (HHA) will be conducted to ensure conformance to 3.2.4.1 and 3.2.4.2, as required by the qualifying activity. The Navy and Marine Corps Public Health Center (NMCPHC) will evaluate the mount using data provided by the manufacturer/distributor to the NMCPHC (see 3.2.4.1 and 6.9).

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The mount assemblies covered in this specification are intended for noise and vibration attenuation. The mounts are primarily for use in naval submarines and surface ships.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Type of mount (see 1.2.1).

c. Type of material to be used in the manufacture of metal parts (see 3.2.1).

d. Protective treatment, if different from that specified on Drawing 803-1385778 (see 3.2.3).

e. Packaging requirements, shipping, and marking (see 5.1 and 6.8).

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of the award of contract, qualified for inclusion in Qualified Products List QPL No. 17508 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and the manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard DC 20376-5160 or emailed to CommandStandards@navy.mil. An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at https://assist.dla.mil.

6.3.1 Additional qualification testing provisions. NAVSEA reserves the right to witness tests and perform any of the tests set forth herein where such testing is deemed necessary to ensure compliance to prescribed requirements of the qualification tests. Qualification testing may be conducted only after NAVSEA has authorized such testing in writing. Copies of SD-6, "Provisions Governing Qualification," are available online at http://quicksearch.dla.mil.

6.3.2 Spring constant values. Spring constant values are to be verified by NAVSEA during qualification (see 6.3.1) and when requested by the mount manufacturer.

6.4 Definitions.

6.4.1 Axial direction. A direction that is parallel to the center bolt axis of the mount. Refer to Drawing 803-1385778.

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6.4.2 Radial direction. A direction that is perpendicular to the center bolt axis of the mount. Refer to Drawing 803-1385778.

6.5 Measurement of test apparatus stiffness. Where practicable, the stiffness of those components of the test apparatus in series with the specimen should be determined experimentally as follows:

a. For test machines capable of such operation, place the machine in the load feedback mode with the displacement transducer on its most sensitive range, and the force transducer on its full scale setting.

b. Bring the platens of the machine either directly in contact or against a stiff steel spacer separating the platens. **Caution: This procedure is not recommended for test machines not equipped with a safety interlock system to prevent the force from exceeding a predetermined value.**

c. The force applied should be sufficient to ensure good resolution from both the force and the displacement instrumentation. Sufficient points should be taken to ensure that the force-deflection relationship is linear and that effects such as non-parallelism between platens do not produce nonlinear results.

d. The stiffness of the test apparatus $K_{apparatus}$ should be determined from the "best fit" straight line through the data points of the force-deflection curve, disregarding anomalous (i.e., abnormal or irregular) points, which may occur at low loads prior to proper seating of the test machine components.

e. Where it is not practical to experimentally determine the stiffness of the test apparatus, the manufacturer's published stiffness for each component in series with the test specimen should be used to calculate the overall stiffness of the test apparatus.

6.6 Suggested formulations and application procedure for oil-resistant coating (see 6.7).

6.6.1 Suggested formulations. The formulations shown below should be blended together just prior to use. The viscosity of the blend should be adjusted to about 650 centipoises by using more or less methyl ethyl ketone in the formulations.

TABLE XVIII. Suggested coating formulations.

	A	B
Prefluxed blend of 70% nitrile rubber with 30 percent polyvinyl chloride	100	100
N330 carbon black	30	30
Stearic acid	1	1
Zinc oxide	5	5
CPB (Dibutyl xanthogen disulfide)	8	---
DBA (dibenzyl amine)	---	8
Sulfur	---	4
Methyl ethyl ketone	656	674
NOTE: Rubber formulations based on parts per hundred rubber (phr).		

6.6.2 Suggested procedure for application of coating to mounts. The sequence of operations shown below should be followed in order to obtain satisfactory performance from the coating:

a. Remove rubber flash from component by trimming or buffing.

b. Sandblast the component, brush off dust.

c. Apply primer in accordance with manufacturer's recommendation to all metal surfaces.

d. Wash the rubber portion of the component with an aqueous solution of Vel detergent, or equal, and rinse with tap water.

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e. Chlorinate the rubber surfaces by immersing each component for 3 minutes in water saturated with chlorine.

f. Rinse mount components with tap water and dry until all traces of water have evaporated.

g. Blend the A solution with the B solution and let stand 1 hour to permit entrapped air to escape.

h. Apply two coats (0.003-inch minimum dry film thickness) of the blended formulations to the entire component, including metal parts, by brushing, dipping, or spraying.

i. Determine the thickness of coating on the rubber element by measuring the thickness of the coated metal plate of the mount (see 4.4.9).

j. Apply additional coats on the metal parts of the mount only to bring the minimum dry film thickness of the coating up to 0.006 inch (on the metal parts only).

k. Allow all coats except the final coat to dry for at least 1 hour before subsequent coats are applied. Allow the final coat to dry for at least 4 days before handling the mount. Alternatively, the final coat should be allowed to dry for at least 4 hours, and then the component placed in an oven at 130±5 °F for 16 hours to cure the coating.

6.7 Disclaimer. The furnishing of information, formulations, and procedures (see 6.6) are solely for assistance in fabrication of the mounts for naval use. The use of this information does not constitute any agreement or obligation by the Department of the Navy to acquire mounts made of these formulae. Also, the use of this information does not guarantee compliance with this specification, nor will it relieve the contractor from having the mounts tested against the applicable test requirements of this specification. Furthermore, the Government does not guarantee that the mounts made using these formulations will conform to the requirements of this specification.

6.8 Marking. Waterproof marking should be provided on interior packages and exterior shipping containers with the following information (see 6.2):

MOUNT TYPE:

DEFENSE SPEC NO: MIL-DTL-17508

DATE OF MFG:

LOT NO:

SHELF LIFE: 7 YEARS FROM DATE OF MFG

Note: The date of manufacture should correspond to the elastomer cure date.

6.9 Toxicity evaluation. The NMCPHC requires sufficient information to permit an HHA of the product. Upon completion of the HHA, a copy will be provided by the NMCPHC to the Government for evaluation. The HHA process is described on the NMCPHC's website, http://www.med.navy.mil/sites/nmcphc/industrial-hygiene/Pages/health-hazard-assessment.aspx.

6.10 Subject term (key word) listing.

Isolation

Rubber

Shock

Vibration

6.11 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issues due to the extent of the changes.

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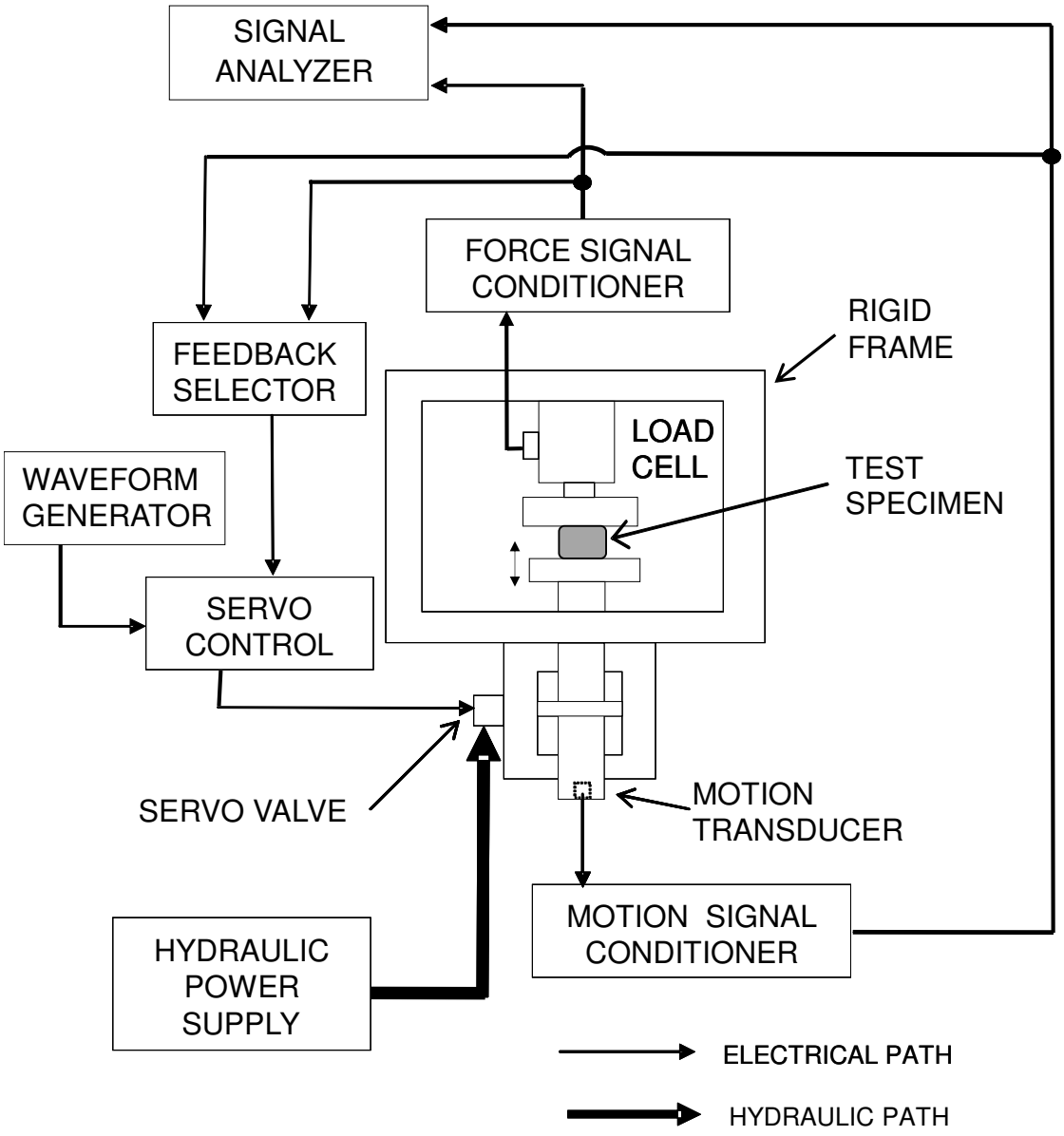


FIGURE 1. Schematic diagram of major components of electrohydraulic closed-loop test machine.

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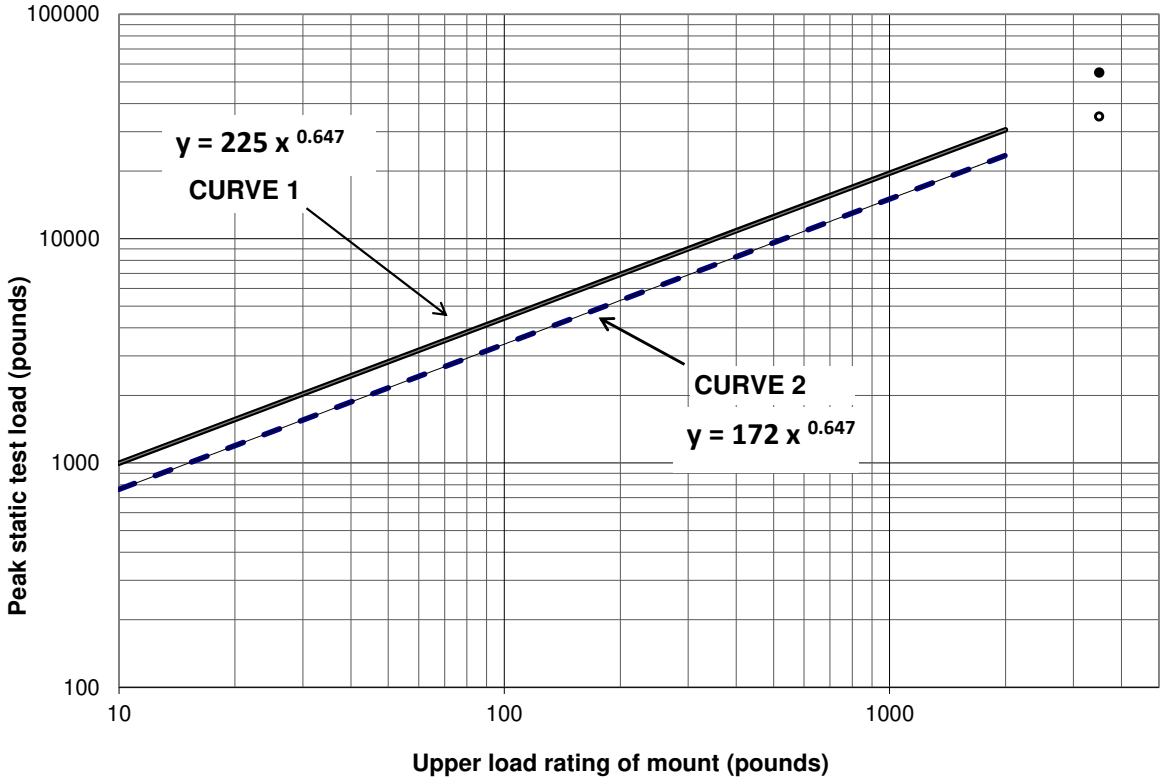


FIGURE 2. Minimum static test strength loads for resilient mountings.

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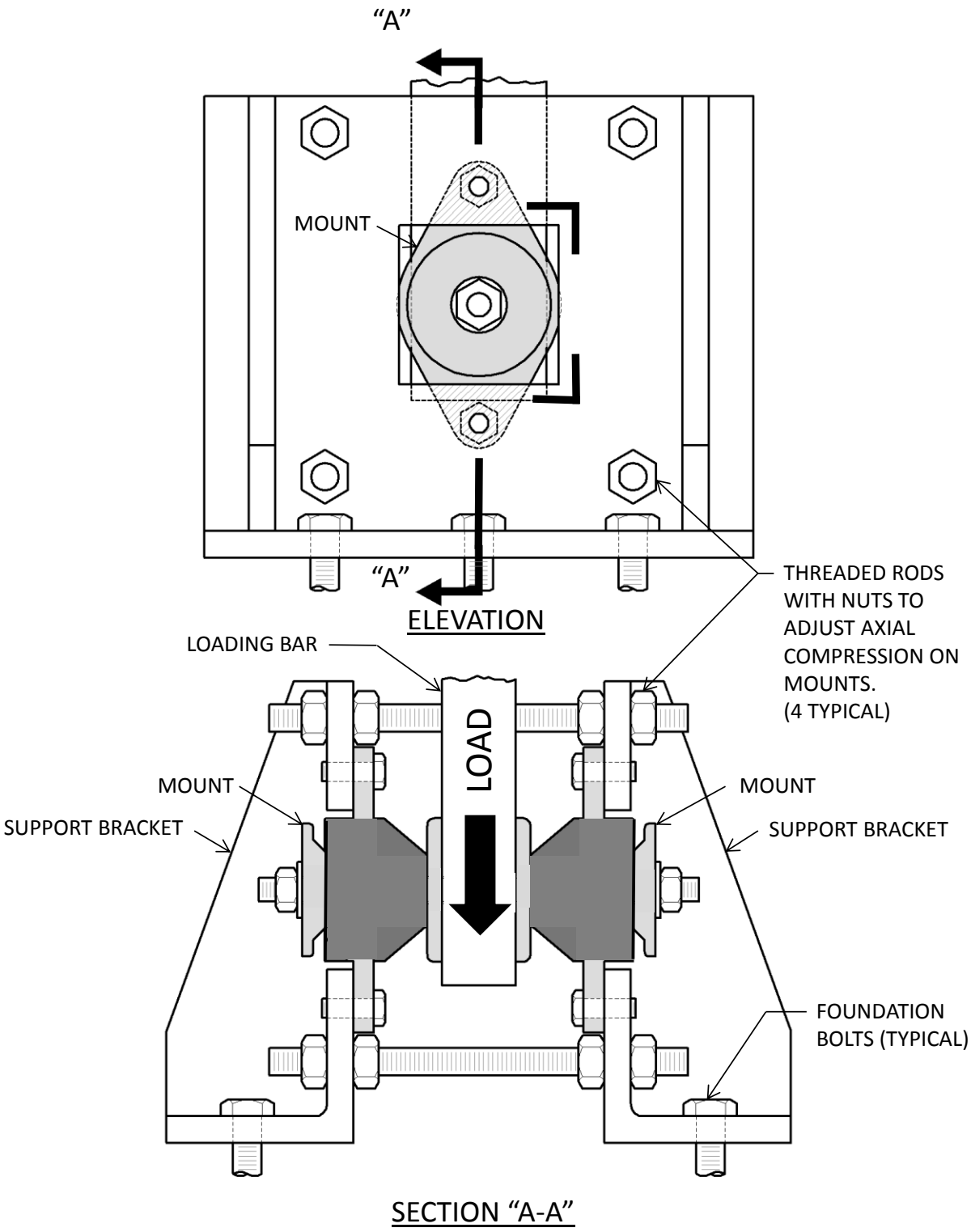


FIGURE 3. Fixture for holding mounts when conducting tests in the radial direction.

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CONCLUDING MATERIAL

Preparing activity:
Navy – SH
(Project 5342-2016-002)

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